A POPULATION-GENETICS PRIMER

Reading Selection in the Genome

How natural selection leaves footprints in human DNA — and how we tell those footprints apart from the tricks of history and chance.

A self-contained companion to three papers:

Vitti, Grossman & Sabeti (2013) · Cheng & Steinrücken (2024) · Mills & Mathieson (2022)

Written for a reader with an undergraduate background in biology and genetics, and no prior population genetics.

How to use this book

This primer exists to make three dense review papers readable. Each was written for specialists, and each assumes a working knowledge of population genetics that a biology degree doesn't usually provide. Rather than have you bounce between a dozen Wikipedia tabs, this book builds the needed background from the ground up, in the order the papers will demand it.

The through-line is a single detective problem: *Given a stretch of DNA, how do we tell whether natural selection has acted on it?* That question turns out to organize the entire field, and it is the exact question you'll bring to genomic regions associated with disease.

> **THE ONE IDEA TO HOLD ONTO**
>
> Selection is never observed directly. It is always **inferred as a deviation from what we would expect if nothing but chance and history were at work**. Every method in this book is therefore one of two things: (a) a way to measure a footprint that selection leaves, or (b) a way to make sure that footprint isn't really population history or a data artifact in disguise.

That framing also explains why your three papers form a natural set. **Vitti et al.** catalogue the footprints and the statistics that measure them. **Cheng & Steinrücken** address the single biggest impostor — population history, or *demography* — and the modern machinery for modelling it. **Mills & Mathieson** supply the humility: the settings where these methods break, and a cautionary tale every practitioner should know. Chapters 1–6 build the concepts; Chapter 7 maps them back onto each paper; the appendices give you a glossary and a practical guide to the databases you'll need.

Chapters are short by design. Each opens with learning objectives and closes with a summary, a list of key terms, and a few questions to test yourself. Worked examples and cautionary cases appear in shaded boxes. You do not need mathematics beyond the idea of an average and a distribution; where an equation appears, it is there to be recognized, not derived.

Contents

CHAPTER 1

The Logic of Population Genetics

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Explain how population genetics differs from the genetics of individuals.
- Name the forces that change allele frequencies and describe what each one does.
- Describe genetic drift and why it is stronger in small populations.
- State the neutral theory and explain why it is the field's null hypothesis.
- Explain why human demographic history makes non-African genomes less diverse.

1.1 From individuals to gene pools

The genetics you learned as an undergraduate mostly asks what a variant *does* inside an organism: this allele of this gene produces this trait. Population genetics zooms out. It treats a population as a **gene pool** — the whole collection of gene copies carried by all its members — and asks how the **frequency** of each variant changes from one generation to the next, and why.

A **variant** (often a single-letter difference in the DNA called a single-nucleotide polymorphism, or **SNP**) that is present on 8% of chromosomes in one population and 40% in another is telling a story about the past. Population genetics is the discipline that learns to read that story. The reason this matters for medicine is direct: disease-associated variants live in populations, and their frequencies, their neighbourhoods, and their histories all carry information about why the disease exists at all.

1.2 The forces that move allele frequencies

Only a handful of processes can change how common a variant is. Learn these five and you have the vocabulary for everything that follows.

Mutation creates brand-new variants. It is the ultimate source of all variation, but it is slow, so it sets the raw material rather than the pace of change.

Genetic drift is random change in frequency caused simply by which individuals happen to reproduce. It is the default engine of the genome and the subject of Section 1.4.

Gene flow (migration) moves variants between populations and tends to make them more alike.

Natural selection is differential survival and reproduction. It is the only force that changes a variant's frequency *because of what the variant does*. Everything in this book is ultimately about detecting its fingerprints against the noise of the other forces.

Recombination is not a force on frequencies at all, but it belongs in this list because it is the great shuffler. Each generation it swaps segments between the chromosome copies you inherited from your two parents, breaking up combinations of nearby variants. As we'll see, recombination is the clock that decides how long selection's footprints survive.

1.3 Hardy–Weinberg: the "nothing is happening" baseline

Before you can detect that *something* is happening, you need a picture of a population in which nothing is. That picture is **Hardy–Weinberg equilibrium**. If a variant has frequency p and its alternative has frequency q (with $p + q = 1$), and if there is no selection, drift, migration, or mutation and mating is random, then the genotype proportions settle to p^2, $2pq$, and q^2 and stay there. The value of Hardy–Weinberg is not that real populations obey it — they rarely do exactly — but that *departures* from it flag that one of the forces is at work. This is your first taste of the field's core move: define the null, then hunt for deviations.

1.4 Genetic drift and effective population size

Imagine a population in which a variant is truly neutral — it makes no difference to survival or reproduction. Its frequency still won't sit still. By pure chance, some carriers will have more offspring than others, and the frequency will wander up and down at random. This aimless wandering is **genetic drift**, and given enough time it drives every neutral variant either to disappear (frequency 0) or to take over completely (frequency 1, called **fixation**).

The crucial property of drift is that **it is stronger in smaller populations**. Flip a fair coin ten times and a 70–30 split is unremarkable; flip it ten million times and you'll land very close to 50–50. Small populations sample their gene pool noisily each generation, so frequencies lurch; large populations sample it finely, so frequencies barely budge.

BOX 1.1 · WORKED INTUITION

Why a jar of marbles predicts the genome

Picture a jar with 10 red and 10 blue marbles. To make the "next generation," draw 20 marbles *with replacement* and let that be the new jar. You will rarely get exactly 10–10; maybe 12–8, maybe 7–13. Repeat, and the counts stagger around until one colour vanishes. Now do the same with 10,000 of each: the proportions hardly move for a very long time. The marbles are gene copies, the draws are reproduction, and the wandering is drift. The single number that captures "how big is the jar, for the purposes of drift" is the effective population size.

That number is the **effective population size**, written $\mathbf{N_e}$. It is almost always much smaller than the actual head-count, because it is dragged down by lean times in a population's past. Humans are a striking example: despite there being billions of us, the long-term human N_e is only around 10,000, a legacy of small ancestral populations and past bottlenecks.

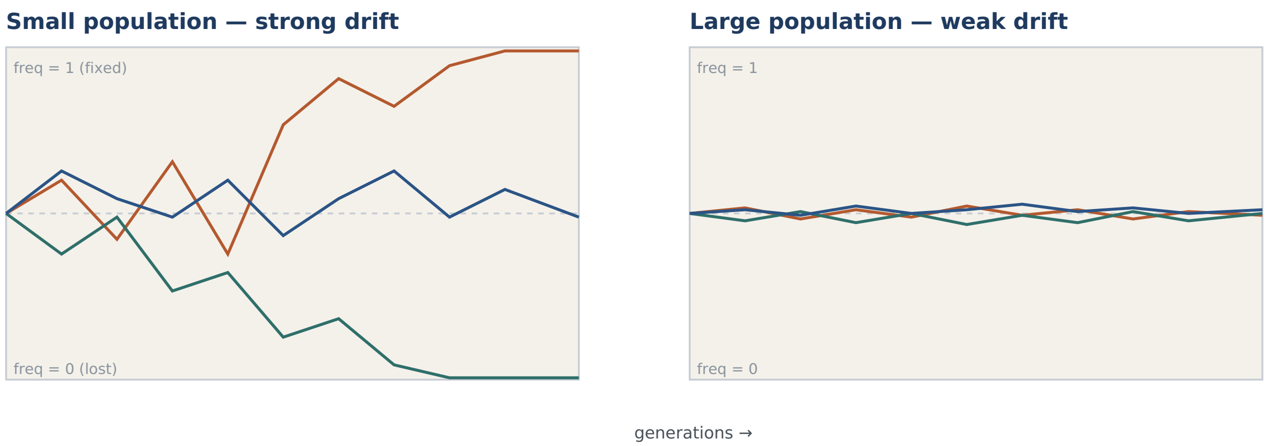

Figure 1.1 Drift depends on population size. Each coloured line is the frequency of one neutral variant over generations. In a small population (left) frequencies swing wildly and some variants are lost or fixed; in a large population (right) the same neutral variants barely move. Drift is the genome's baseline hum — the thing selection must be distinguished from.

N_e has a second consequence you will meet constantly. As modern humans expanded out of Africa, the founding non-African groups passed through a population **bottleneck** — a sharp, temporary reduction in size. Bottlenecks are drift on overdrive: they purge variation. As a result, non-African populations today carry *less* genetic diversity and *longer* stretches of correlated DNA, while African-ancestry populations retained more diversity and shorter correlated blocks. Keep this fact close: it is why analyses that include diverse and especially African-ancestry samples have sharper resolution, a theme that will return in every later chapter.

1.5 The neutral theory: the field's null hypothesis

In 1968 Motoo Kimura proposed the **neutral theory of molecular evolution**: the claim that the *great majority* of genetic variation within and between species is selectively neutral and is governed by drift, not by selection. The neutral theory was controversial as biology, but its role in this book is not as a belief — it is as a **null hypothesis**. It gives us a precise, mathematical prediction for what a genome shaped by mutation, drift, and demography *alone* should look like. Selection is then detected exactly where the real data refuse to match that prediction.

> **THE MOVE THAT DEFINES THE FIELD**
>
> Model a world with no selection (mutation + drift + demography). Ask what patterns that world produces. Then find the places in a real genome where the pattern is **too extreme to be that world**. Those places are your candidates for selection. Everything technical from here on is a refinement of this one idea.

SUMMARY

- Population genetics studies how allele frequencies change in a gene pool over generations.
- Five processes matter: mutation, drift, gene flow, selection, and (as a shuffler) recombination.
- Genetic drift is random frequency change; it is stronger when the effective population size N_e is small.
- Human N_e is ~10,000; the out-of-Africa bottleneck left non-African genomes less diverse and more correlated.
- The neutral theory supplies the null hypothesis: selection is inferred as a deviation from a drift-only expectation.

KEY TERMS

gene pool · allele frequency · SNP · mutation · genetic drift · gene flow · natural selection · recombination · Hardy–Weinberg equilibrium · fixation · effective population size (N_e) · bottleneck · neutral theory · null hypothesis

CHECK YOUR UNDERSTANDING

1. Two islands have the same census population but very different N_e. Which will show more drift, and what feature of its history could explain the low N_e?
2. Why is the neutral theory useful even to a researcher who is certain selection is important?
3. Predict, in one sentence each, how an out-of-Africa bottleneck should affect (a) diversity and (b) the length of correlated DNA blocks in non-African genomes.

CHAPTER 2

How Selection Leaves Footprints

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Distinguish positive, purifying, and balancing selection, with an example of each.
- Define linkage disequilibrium and explain genetic hitchhiking.
- List the four things a selective sweep does to the DNA around a favoured variant.
- Explain why recombination makes recent sweeps easier to detect than ancient ones.
- Contrast hard and soft sweeps.

2.1 Three modes of selection

Positive (directional) selection favours a beneficial variant and drives its frequency up. This is the mode most detection methods hunt, because a rising variant disturbs its surroundings in tell-tale ways.

Purifying (negative) selection removes harmful variants. It is by far the most common form of selection and it is why functionally important stretches of the genome are **conserved** across species. But because it mostly *prevents* variation from ever becoming common, its signature is quiet — an absence rather than a splash.

Balancing selection actively *maintains* two or more variants at intermediate frequencies rather than letting one win. The textbook case is the sickle-cell variant, held in the population because carrying one copy protects against malaria even though two copies cause disease (Box 2.1). The immune **HLA** genes are the other classic example: it pays to keep a diverse arsenal of variants so the population can recognize many pathogens. Balancing selection at immune genes is a pattern worth remembering; it recurs wherever host–pathogen conflict shapes the genome.

BOX 2.1 · CASE STUDY

Sickle cell: selection that preserves a “harmful” allele

The sickle-cell variant of the haemoglobin gene causes anaemia in people who inherit two copies. Left to selection against the disease, it should be rare everywhere. Yet it reaches appreciable frequencies across malarial regions of Africa, the Mediterranean, and South Asia. The reason is **heterozygote advantage**: people with one copy are substantially protected from severe malaria. Selection therefore pushes the variant *up* (via the protected carriers) and *down* (via the sick homozygotes) at once, and it settles at an intermediate frequency. This is balancing selection, and it is a clean reminder that “disease allele” and “selected-against” are not synonyms — a point with obvious resonance for the genetics of complex disease.

2.2 Linkage, hitchhiking, and the reason selection is visible

Here is the mechanism that makes selection detectable at all. DNA is inherited in blocks, not one variant at a time. Because of that, nearby variants tend to travel together, and the statistical signature of “these variants co-occur more often than chance predicts” is called **linkage disequilibrium**, or **LD**. Two variants in strong LD are, in effect, a package deal on the chromosomes that carry them.

Now suppose selection grabs one variant and drives it upward. Every chromosome that carries the favoured variant also carries whatever else happened to sit nearby on that same DNA segment — and all of it rises in frequency together. Neighbouring variants get a free ride purely by association. This is **genetic hitchhiking**, and it is why a single favoured site leaves a footprint spread across a whole neighbourhood of the genome that we can actually measure.

2.3 The selective sweep and its four consequences

When a favoured variant rises from rare to common, dragging its neighbourhood along, the event is called a **selective sweep**. A sweep leaves four linked consequences, and essentially every statistic in Chapter 4 is a way of measuring one of them.

(1) Reduced local diversity. Because the surviving chromosomes in the region are mostly copies of the one lucky chromosome the favoured variant arose on, variation near the site collapses — everyone looks alike there.

(2) Long, identical haplotypes. A **haplotype** is a specific combination of variants inherited together on one stretch of chromosome. Right after a sweep, many people share a long, unbroken haplotype, because recombination has not yet had time to chop it up.

(3) A distorted frequency pattern. The sweep leaves an unusual pile-up of variants at particular frequencies — a distortion of the site frequency spectrum you'll meet in Chapter 3.

(4) Differences between populations. If the sweep happened in one population but not another, the region will look strikingly different between them — the basis of the “cross-population” tests.

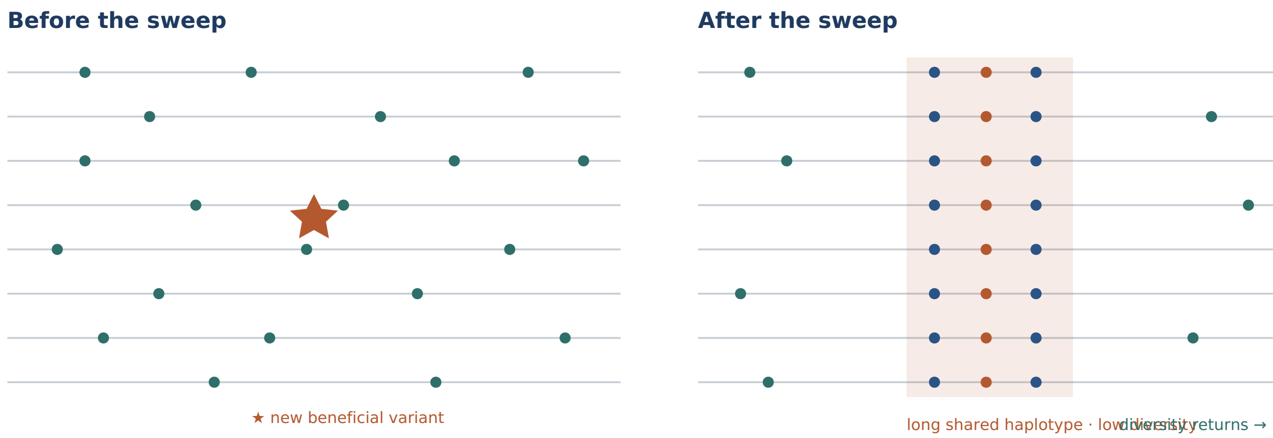

Figure 2.1 A selective sweep. Each horizontal line is one chromosome's copy of a genomic region; dots are variants. **Before**, the region is diverse and a new beneficial variant (★) sits on a single chromosome. **After** the sweep, that variant is on nearly every chromosome, and it has dragged its immediate neighbours (blue) along, so the core region is a single long, identical, low-diversity haplotype. Diversity survives only at the flanks, where recombination has separated variants from the favoured site.

2.4 Recombination is the clock

A sweep's most dramatic footprint — the long shared haplotype — is temporary. Every generation, recombination snips a little more off the block, trading the long identical stretch for shorter and shorter ones. So the length of the shared haplotype is a **clock**: a long block means the sweep was *recent* (little time to erode), whereas a sweep from tens of thousands of years ago has had its haplotype shattered and leaves only a faint frequency signature behind.

This single fact explains a puzzle you'll notice in Chapter 4: why are there so many different statistics? Because they detect selection over *different timescales*. Haplotype-based methods see recent sweeps whose long blocks still survive; frequency-based methods can reach further back, to sweeps whose haplotypes are long gone but whose frequency distortions linger.

2.5 Hard and soft sweeps

The clean picture in Figure 2.1 is a **hard sweep**: a single new mutation is favoured and carries one specific haplotype to high frequency. Reality is often messier. In a **soft sweep**, selection acts on variation that was *already present* in the population (called **standing variation**), or the favoured variant arose more than once on different chromosomal backgrounds. Because several haplotypes rise together, a soft sweep smears the footprint — diversity doesn't collapse as sharply, and no single long haplotype dominates. Soft sweeps are consequently much harder to detect, and their prevalence is one reason the field has grown cautious about how many sweeps it can confidently claim.

SUMMARY

- Positive selection raises beneficial variants; purifying selection removes harmful ones; balancing selection maintains multiple variants (sickle cell, HLA).
- Nearby variants travel together (linkage disequilibrium); a favoured variant drags its neighbours up with it (hitchhiking).
- A selective sweep reduces local diversity, creates long identical haplotypes, distorts the frequency spectrum, and differentiates populations.
- Recombination erodes the shared haplotype over time, so haplotype length dates a sweep — long block, recent sweep.
- Hard sweeps (one new mutation) leave sharp footprints; soft sweeps (standing or repeated variation) are muddier and harder to find.

KEY TERMS

positive/directional selection · purifying selection · conservation · balancing selection · heterozygote advantage · HLA · linkage disequilibrium (LD) · genetic hitchhiking · selective sweep · haplotype · hard sweep · soft sweep · standing variation

CHECK YOUR UNDERSTANDING

1. Why does purifying selection, despite being the most common form of selection, leave the least conspicuous footprint?
2. Two sweeps reached the same final frequency, but one left a long shared haplotype and the other did not. What differs between them?
3. Explain why a soft sweep is harder to detect than a hard sweep, in terms of what happens to local diversity.

CHAPTER 3

Two Views of the Data

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Build a site frequency spectrum by hand from a small dataset, in two counting passes.
- Define the site frequency spectrum and read its shape.
- Say how the meaning of “rare” and “common” depends on sample size.
- Explain why distinguishing ancestral from derived alleles matters.
- Describe how sweeps and balancing selection bend the spectrum in opposite directions.
- Explain when the SFS is computed genome-wide versus in local windows.
- Explain what haplotype structure and LD decay reveal that the spectrum does not.

Nearly every method for detecting selection reads one of two summaries of a stretch of genome. Master these two “views” and the statistics in Chapter 4 stop being a list of names and become obvious variations on a theme.

3.1 The site frequency spectrum

Take a genomic region and, for every variable site in it, count how many chromosomes in your sample carry the variant. Then make a histogram: how many sites have their variant at low frequency, how many at intermediate frequency, how many at high frequency. That histogram is the **site frequency spectrum**, or **SFS**. It is the workhorse summary of population genetics because the neutral theory predicts its shape precisely, and both selection and demography bend it in characteristic ways.

The fastest way to make the SFS concrete is to build one by hand from a tiny dataset. It takes exactly **two counting passes**, and the second pass — called **pooling** — is where the idea usually clicks.

Suppose we sequence the same short stretch of DNA from just **five chromosomes** (five people, one chromosome each). Most positions are identical in all five and are ignored; we keep only the **variable sites**, and at each we write **0** for the original (ancestral) letter and **1** for the new variant:

	s1	s2	s3	s4	s5	s6	s7	s8
Chromosome 1	0	1	0	0	0	1	1	0
Chromosome 2	0	0	0	1	1	1	0	0
Chromosome 3	1	0	0	1	0	1	1	0
Chromosome 4	0	1	0	1	0	1	0	1
Chromosome 5	0	0	1	0	0	0	0	0
Variant copies	1	2	1	3	1	4	2	1

Pass 1 — count the variant at each site (one number per site). Go down each column and count the 1s; that is the shaded bottom row. Site 1 has the variant in one chromosome, site 4 in three, site 6 in four. Reading across, the eight sites give eight numbers: **1, 2, 1, 3, 1, 4, 2, 1**. Now set the table aside — everything that follows uses only this list of eight numbers.

Pass 2 — pooling: count how often each number appears in that list. This step is no longer about DNA; it is ordinary tallying. How many sites have their variant in exactly one chromosome? Sites 1, 3, 5, and 8 — so **four**. In exactly two? Sites 2 and 7 — **two**. In three? One site. In four? One site. (Check: 4 + 2 + 1 + 1 = 8, every site accounted for.) The key mental shift is that in Pass 2 the individual sites *lose their identity*: we stop asking *which* site was rare and ask only *how many* sites were rare. Those four singleton sites dissolve into one fact — “four sites had a lone variant.” That dissolving is the pooling, and the resulting tally is the SFS.

Figure 3.1 An SFS built by hand. The tally from the five-chromosome table, drawn as a histogram: four sites carry a singleton variant (one copy), two sites carry two copies, and one site each carries three and four copies. Every bar counts *sites*, not people; the horizontal axis is how common the variant is. Even this toy already slopes downward — many rare, few common — the natural neutral shape explained next.

So the axes mean exactly this: the **horizontal axis is how common the variant is** (how many chromosomes carry it — rare on the left, common on the right), and the **vertical axis is how many sites are that common**. The single most common misconception is that the SFS describes one variant; it does not. It is a **census of every variant you**

found, sorted into bins by how common each one is — the genetic equivalent of turning a list of exam scores into a grade distribution, where the “score” is a variant's frequency.

How rare is “rare,” how common is “common”?

The bins are set by your sample size. In a sample of n chromosomes, the leftmost bin is a variant seen in exactly **one** chromosome — the rarest thing possible, called a **singleton** — and the rightmost useful bin is a variant seen in $n - 1$ chromosomes (present in all but one). With five chromosomes the bins run only 1–4, which is why our histogram is chunky. Real studies use far larger samples, and this changes the texture completely. Sequence 1,000 people — that is 2,000 chromosomes — and the bins run from 1 all the way to 1,999, so the histogram becomes a smooth curve. “Rare” then means something striking: a singleton is one copy out of 2,000, and in a biobank of hundreds of thousands of people it is one copy in hundreds of thousands — a variant so young or so suppressed that almost no one carries it. “Common” means the variant is carried by a large fraction of everyone sampled, often tens of percent. The smooth curve is the shape you actually read for selection; our five-chromosome toy is the same idea shrunk down to where you can count it by hand.

Under pure neutral drift, the SFS has a familiar tilt: there are many rare variants and progressively fewer common ones (mathematically, the number of sites at frequency i falls off roughly as $1/i$). Deviations from that expected curve are the signal.

To read the spectrum fully you must know, for each site, which version is the original and which is new. The original is the **ancestral allele** and the new one is the **derived allele**; you assign them by comparing to an **outgroup** species such as chimpanzee. A spectrum that keeps this distinction is **unfolded**; one that cannot (because the ancestral state is unknown) is **folded**. The distinction matters because some selection signatures — notably an excess of *high-frequency derived* alleles left by hitchhiking — are invisible unless you know which allele is derived.

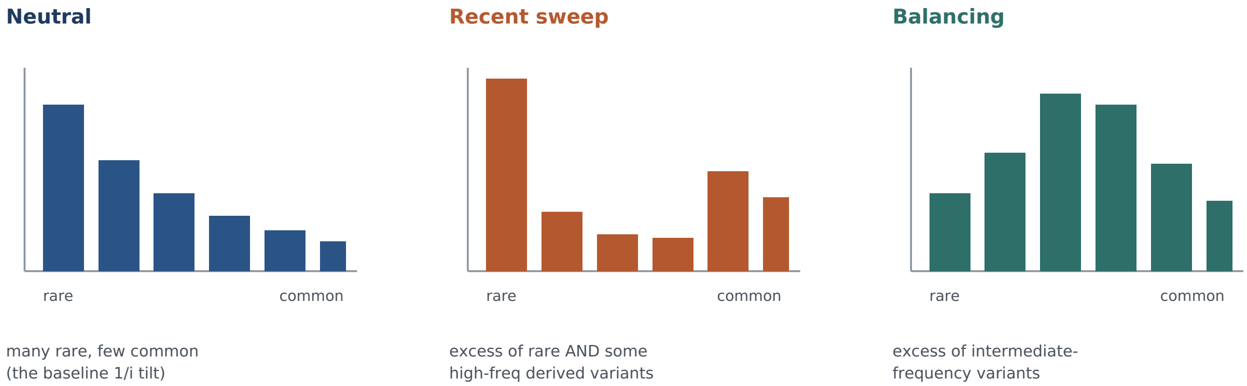

Figure 3.2 Three site frequency spectra. Bars show how many variant sites fall in each frequency class, from rare (left) to common (right). Neutral drift gives the tilted baseline. A recent sweep inflates the rare class (young variants on the swept background) and can add a bump of high-frequency derived variants from hitchhiking. Balancing selection does the opposite, piling variants at intermediate frequency. Statistics like Tajima's D are simply numerical ways of asking which of these shapes a region has.

BOX 3.1 · SCOPE

One region, or the whole genome?

A natural question once you can build an SFS: is it computed from the whole genome at once, or region by region? Both are done, and *which* you choose depends on what you are studying. A **genome-wide SFS** pools variants from across the entire genome into one big tally. Because demography — bottlenecks, expansions, migrations — affects the whole genome roughly uniformly, this whole-genome spectrum is what you use to *infer population history*. A **local (windowed) SFS** pools only the sites inside one region — a single gene, or a sliding window a few hundred kilobases wide — because *selection acts locally*: a sweep distorts the spectrum in its own window while the rest of the genome looks normal. In practice a selection scan slides a window along each chromosome and computes a fresh SFS (and statistics such as Tajima's D) in every window, then asks which windows are outliers against the genome-wide backdrop. Same two-pass counting either way — the only difference is how much of the genome you pour into the tally. This split between whole-genome and local views is the seed of the central idea in Chapter 5.

BOX 3.2 · HOW TO READ A SPECTRUM

Left-heavy vs. middle-heavy

A quick heuristic: if a region's spectrum is *more left-heavy* than neutral (too many rare variants), suspect a recent sweep or a population that recently grew. If it is *more middle-heavy* than neutral (too many intermediate-frequency variants), suspect balancing selection or a population that recently shrank. Notice that each selection signature has a demographic look-alike — the central problem of Chapter 5, previewed here.

3.2 Haplotype structure: what the spectrum can't see

The SFS throws away one crucial thing: which variants sit together on the same chromosome. That information — the **haplotype structure** — is the second view. Its key readout is how quickly **LD decays** with distance: as you move away from a given site, how fast do the neighbourly associations break down? Around a recent sweep, LD stays high over an unusually long distance, because the swept haplotype is long and shared by many people. The spectrum can miss a very recent sweep that hasn't yet distorted allele frequencies much; the haplotype view catches it, because the tell-tale long block is already there. The two views are complementary, and the strongest evidence for selection is when both point the same way.

SUMMARY

- The site frequency spectrum (SFS) is the histogram of how common variants are across a region; the neutral theory predicts its shape.
- Build one in two passes: count the variant at each site, then pool those counts by tallying how many sites fall in each frequency class.
- The frequency classes run from a singleton (one copy) up to $n-1$ copies; larger samples make the spectrum a smooth curve.
- The SFS can be computed genome-wide (to study population history) or in local windows (to detect selection).
- Sweeps make the SFS more left-heavy (excess rare, plus high-frequency derived variants); balancing selection makes it more middle-heavy.
- Reading high-frequency-derived signals requires distinguishing ancestral from derived alleles using an outgroup (an unfolded spectrum).
- Haplotype structure — how fast LD decays — adds what the SFS omits and is especially sensitive to very recent sweeps.

KEY TERMS

site frequency spectrum (SFS) · pooling · singleton · genome-wide vs. windowed SFS · ancestral allele · derived allele · outgroup · folded/unfolded spectrum · haplotype structure · LD decay

CHECK YOUR UNDERSTANDING

1. Given the eight per-site counts 2, 1, 1, 3, 1, 2, 1, 1, perform the pooling pass and state the resulting spectrum (how many sites in each frequency class).
2. In one sentence each, say when you would compute a genome-wide SFS versus a local windowed SFS, and why.
3. A region has far more intermediate-frequency variants than the neutral expectation. Name the two very different explanations you must now distinguish.
4. Why can't a folded spectrum reveal an excess of high-frequency *derived* alleles?
5. Give one situation in which the haplotype view detects a sweep that the frequency-spectrum view would miss.

CHAPTER 4

The Statistics, Decoded

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Sort the major selection statistics by which footprint they measure.
- Interpret the sign of Tajima's D.
- Explain what iHS, FST, XP-EHH, and a composite scan each detect.
- State the golden rule that governs all of them.

The acronyms in the selection literature are not a random pile. Each one is a ruler laid against a footprint from Chapters 2–3. Grouped by footprint, they are easy to keep straight.

4.1 Frequency-spectrum tests (within one population)

Tajima's D is the workhorse. It compares two ways of measuring genetic diversity that weight rare and intermediate variants differently; the direction in which they disagree reports the shape of the spectrum. A *negative* D means an excess of rare variants (a recent sweep — or a population expansion). A *positive* D means an excess of intermediate-frequency variants (balancing selection — or a population contraction). **Fay and Wu's H** targets the specific excess of *high-frequency derived* alleles that hitchhiking produces, so it is more sweep-specific. **Fu and Li's** statistics lean on the very rarest variants. All three read the SFS; they differ in which part of it they listen to.

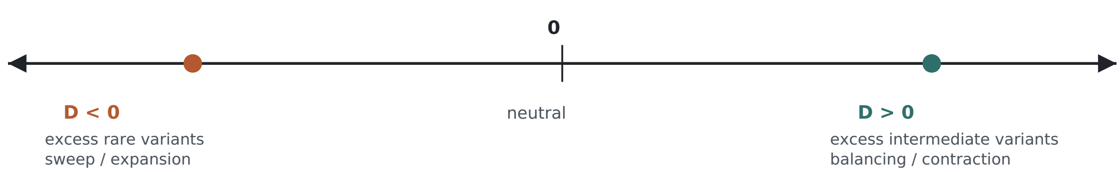

Figure 4.1 Reading Tajima's D. The sign points to the spectrum's shape, but each interpretation has a demographic twin (in parentheses) that a single locus cannot rule out. This ambiguity is exactly why genome-wide context and demographic modelling are required.

4.2 Cross-species tests (selection on genes, deep time)

Two classic tests compare variation *within* a species to divergence *between* species. The **McDonald–Kreitman (MK) test** asks whether a gene has more between-species differences of the kind that change proteins than neutrality predicts — a signature of recurrent positive selection over long timescales. The **HKA test** compares polymorphism and divergence across regions. These operate on a much deeper time axis than the sweep statistics and are aimed at genes, not recent events.

4.3 Haplotype tests (recent sweeps)

EHH (extended haplotype homozygosity) measures how far an identical haplotype extends around a chosen “core” variant — in effect, how slowly LD decays on that background. **iHS** (integrated haplotype score) uses EHH to flag variants sitting on suspiciously long haplotypes *within* a single population; it excels at *incomplete* or still-ongoing sweeps, where the favoured variant hasn't yet reached fixation. **nSL** is a closely related refinement. These are the methods that read the haplotype view directly.

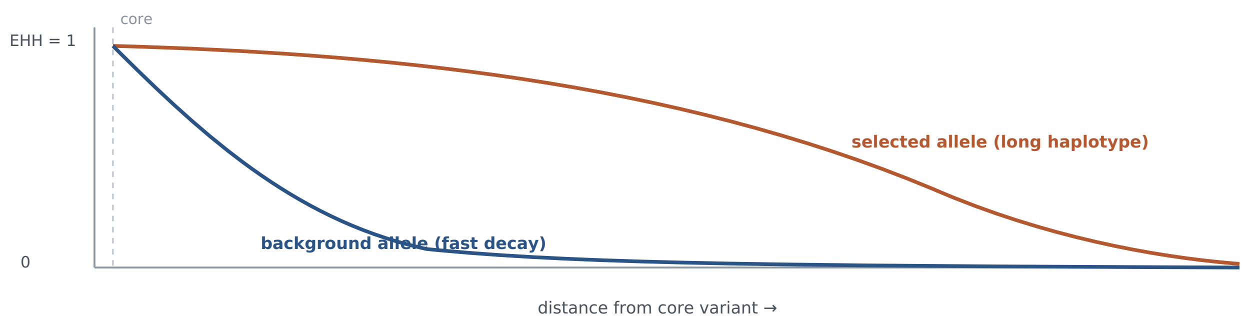

Figure 4.2 Extended haplotype homozygosity (EHH). Moving away from a core variant, the chance that two chromosomes still match falls off. On a recently selected allele (rust) it falls slowly — the swept haplotype is long. On an ordinary background allele (blue) it falls quickly. iHS turns the gap between these two curves into a single number.

4.4 Population-differentiation tests (local adaptation)

When selection acts in one population and not another, the region diverges between them. $\mathbf{F_{ST}}$ is the fundamental measure: it quantifies how much of the total variation at a site is due to differences *between* populations. High F_{ST} means a variant common in one group and rare in another. **ΔDAF** is simply the difference in derived-allele frequency between two populations. **PBS** (the population branch statistic) uses three populations to work out *which lineage* the change happened on. **XP-EHH** and **XP-CLR** are the cross-population versions of the haplotype and composite tests — a long shared haplotype or a sweep-shaped distortion present in one population but absent in another.

Figure 4.3 What F_{ST} measures. Bars show a variant's frequency in two populations. Large between-population differences (left) give high F_{ST} and flag possible local adaptation; near-equal frequencies (right) give low F_{ST}. Population history alone can inflate F_{ST}, so a high value is a lead, not a verdict.

4.5 Composite and machine-learning scans

Because each single statistic is noisy, modern scans combine them. **CLR / SweepFinder** computes the likelihood that a local spectrum distortion is a sweep rather than neutral. **CMS** combines several statistics to sharpen and localize a signal. Machine-learning classifiers such as **S/HIC** go further, taking dozens of summary statistics as input and learning to separate sweeps from neutral-plus-demographic look-alikes. These composite methods are more powerful, but they inherit a dependency we address next: they are only as trustworthy as the demographic model used to define “neutral.”

Statistic	Footprint it reads	Best at
Tajima's D	Shape of the SFS	Sweeps (D<0) vs balancing (D>0), older signals
Fay & Wu's H	High-freq derived alleles	Sweep-specific frequency signal
iHS / nSL	Haplotype length (1 pop)	Recent, incomplete sweeps
F_{ST} / ΔDAF / PBS	Between-population difference	Local adaptation
XP-EHH / XP-CLR	Cross-population haplotype/ spectrum	Sweep in one population, not another
MK / HKA	Within- vs between-species	Recurrent selection on genes, deep time
CLR / CMS / S/HIC	Many footprints combined	Power and localization; need a demographic null

THE GOLDEN RULE

No single statistic is definitive; each detects selection over a characteristic **timescale**; and **every one of them can be mimicked by population history**. Convergent signals from several statistics, interpreted against a genome-wide background and a demographic model, are what make a case — not a lone outlier.

SUMMARY

- Frequency-spectrum tests (Tajima's D, Fay & Wu's H, Fu & Li) read the SFS within one population.
- Haplotype tests (EHH, iHS, nSL) read haplotype length and catch recent, incomplete sweeps.
- Differentiation tests (F_{ST}, ΔDAF, PBS, XP-EHH, XP-CLR) compare populations and flag local adaptation.
- Cross-species tests (MK, HKA) detect recurrent selection on genes over deep time.
- Composite and machine-learning scans combine statistics for power but depend on a demographic null.

KEY TERMS

Tajima's D · Fay & Wu's H · Fu & Li · McDonald–Kreitman test · HKA test · EHH · iHS · nSL · F_{ST} · ΔDAF · PBS · XP-EHH · XP-CLR · SweepFinder/CLR · CMS · S/HIC

CHECK YOUR UNDERSTANDING

1. A locus has strongly negative Tajima's D but the whole genome of that population also skews negative. What non-selective explanation must you consider?
2. Which statistic would you reach for first to detect a sweep that is recent and still incomplete, and why?
3. Why is a high F_{ST} value a lead rather than proof of local adaptation?

CHAPTER 5

The Great Impostor: Demography

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Explain why population history mimics the footprints of selection.
- Describe the two strategies used to separate selection from demography.
- Explain, in words, what the coalescent is and why tree shape is informative.
- Name the kinds of tools used to infer demography and reconstruct genealogies.

This is the chapter that makes the field hard, and it is the heart of Cheng & Steinrücken. The problem is stated in one sentence: **population history distorts the genome in the very same ways selection does.**

5.1 Why demography is an impostor

A population **bottleneck** purges diversity and lengthens haplotypes — but across the *entire* genome, not just at one site. That is the same signature a local sweep makes, only everywhere at once. A recent population **expansion** floods the genome with new, rare variants, mimicking the excess-of-rare-variants footprint of a sweep and pushing Tajima's D negative genome-wide. Population **structure** and **admixture** (mixing of previously separated groups) inflate F_{ST} and create long-range LD that looks like linked selection. Run any single statistic naively across a real human genome and label every outlier “selection,” and you will mostly have rediscovered our species' migrations.

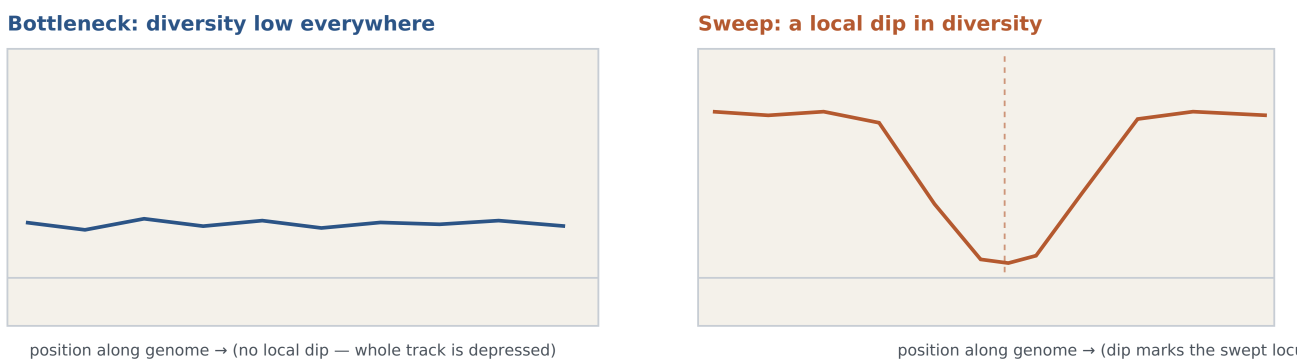

Figure 5.1 Local outlier vs. genome-wide background. Diversity plotted along the genome. A bottleneck lowers diversity *uniformly* (left); a sweep carves a *local* trough against an otherwise normal background (right). Because demography acts genome-wide and selection acts locally, the escape from the impostor is to judge each region against the rest of its own genome.

5.2 Escape route 1: local outliers against the genome-wide background

The first defence follows directly from Figure 5.1. Demography leaves a signature shared by the whole genome; a true sweep is a *local* anomaly rising above that shared baseline. So the right question is never “is this region unusual in absolute terms?” but “is it unusual *relative to the rest of this genome, which shares the same history*?” This is why selection scans hunt for outliers, and why comparing candidate regions to carefully **matched control regions** is non-negotiable. It is also why, when researchers ask whether the loci for a given disease are *collectively* enriched for selection signals against a matched background, the answer is often no — the appropriate baseline has already absorbed the demographic signal.

5.3 Escape route 2: model the demography explicitly

The second, more powerful defence is to build a mathematical model of the population's history — its past sizes, splits, and migrations — then simulate genomes under that history *with no selection*, and ask whether the observed signal exceeds anything the model produces. If your candidate sweep is stronger than 99.9% of simulated neutral regions from a realistic demography, you have a case. This is the machinery Cheng & Steinrücken survey, and it rests on one central idea.

5.4 The coalescent: thinking backward in time

The **coalescent** is the mathematical backbone of modern population genetics. Instead of simulating evolution forward from ancestors to descendants, it runs time *backward*: pick any set of present-day DNA sequences and trace their lineages back until they merge — **coalesce** — into shared ancestors, eventually reaching a single common ancestor. The result is a genealogical tree, and the *shape* of that tree encodes the population's history and any selection.

Why tree shape? Because different histories make lineages coalesce at different times. Under a sweep, the surviving lineages all descend from the one chromosome the favoured variant arose on, so they coalesce suddenly and recently, giving a “star-like” tree with a burst of recent branching. A bottleneck squeezes coalescence times too. A balancing-selection locus does the opposite: it keeps deep, ancient branches alive. Learning to picture genealogies is the single biggest conceptual unlock in the modern literature, because every summary statistic you have met is, underneath, a fingerprint of tree shape.

Figure 5.2 Selection reshapes genealogies. A neutral genealogy (left) has deep, balanced branches. A sweep (right) forces the sampled lineages to a recent common ancestor, producing a shallow, star-like tree. Modern methods reconstruct these trees directly from data and read selection off their shape.

5.5 The modern toolkit

Cheng & Steinrücken tour the tools that put these ideas to work. Three families are worth recognizing by name:

Demographic-inference methods estimate past population sizes and splits from data. Some read the SFS (tools named **∂a∂i**, **moments**, and **fastsimcoal2**); others, the coalescent-HMM methods **PSMC** and **MSMC**, reconstruct a population's size through time from as little as a single genome.

Ancestral recombination graphs (ARGs) are the frontier. An ARG is the full set of genealogies along a chromosome, accounting for how recombination changes the tree from place to place. New tools (**Relate**, **tsinfer/tsdate**) reconstruct ARGs at genome scale, which lets you estimate the **age of an allele** and read selection off tree shape directly (as the method **CLUES** does). This is where the field is heading, and it is why Cheng & Steinrücken devote real space to it.

Machine-learning methods (e.g., **S/HIC**, **diploS/HIC**) treat selection detection as a classification problem, training on simulated genomes to tell sweeps apart from neutral-plus-demographic scenarios. They are powerful precisely to the degree that the training simulations capture the true demography — the same dependency, restated.

BOX 5.1 · CAUTIONARY WORKED EXAMPLE

The bottleneck that looked like adaptation

Suppose you scan a non-African population and find hundreds of regions with reduced diversity and long haplotypes. Exciting — hundreds of sweeps? Before celebrating, recall the out-of-Africa bottleneck (Chapter 1): it reduced diversity and lengthened haplotypes across the *whole* genome. Much of your “signal” is that shared demographic backdrop. Only the regions that stand out *above* the genome-wide bottleneck signature, and that survive comparison to neutral simulations under a realistic demographic model, are credible candidates. This is Chapter 5 in a single paragraph, and it is the mistake the whole modern toolkit exists to prevent.

SUMMARY

- Bottlenecks, expansions, structure, and admixture distort the genome in the same ways selection does — the central confounder.
- Because demography acts genome-wide and selection acts locally, judging regions against a matched genome-wide background separates them.
- The stronger defence is to model demography explicitly and test observed signals against neutral simulations from that model.
- The coalescent runs time backward; genealogical tree shape encodes both history and selection.
- Tools span demographic inference (∂a∂i, PSMC/MSMC), ARG reconstruction (Relate, tsinfer), and machine learning (S/HIC).

KEY TERMS

demography · bottleneck · expansion · population structure · admixture · matched control regions · coalescent · coalescence · genealogy/tree shape · demographic inference · ancestral recombination graph (ARG) · allele age · PSMC/MSMC · Relate/tsinfer · S/HIC

CHECK YOUR UNDERSTANDING

1. Explain in one sentence why a population expansion can produce a genome-wide negative Tajima's D with no selection at all.
2. Why does comparing a candidate region to the rest of the same genome help control for demography?
3. What does the star-like shape of a genealogy under a sweep tell you about when the sampled lineages last shared an ancestor?

CHAPTER 6

The Hardest Case: Recent and Polygenic Selection

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Explain why selection on a polygenic trait is nearly invisible to sweep statistics.
- Describe how polygenic-selection tests use GWAS effect sizes.
- Recount the height-selection episode and the lesson it taught the field.
- Explain why ancient DNA offers a cleaner test of recent selection.

Mills & Mathieson is the reality check of your three papers. It concentrates on the two settings where everything in Chapters 4–5 grows shakiest: selection that is very *recent*, and selection on traits that are *polygenic*.

6.1 Why polygenic selection hides

Most human traits and common diseases are **polygenic**: shaped by hundreds or thousands of variants, each nudging the trait by a tiny amount. Natural selection on such a trait does not sweep any single variant to high frequency. Instead it shifts *many* variant frequencies by a hair each — raising the trait-increasing alleles a little, lowering the others. No individual locus shows the classic sweep footprint, so the haplotype and spectrum statistics of Chapter 4 are nearly blind to it. To catch polygenic selection you must aggregate the whole set of trait-associated variants and look for a *coordinated* shift.

6.2 Polygenic-selection tests and the singleton density score

The standard approach borrows the results of a **genome-wide association study (GWAS)**, which estimates each variant's **effect size** on a trait. A **polygenic-selection test** then asks whether the trait-*raising* alleles have, as a group, drifted systematically up or down in frequency — more than chance allows. A complementary tool, the **singleton density score (SDS)**, reads very recent frequency change (the last few thousand years) from the local density of the rarest variants around an allele: a young, recently favoured allele has not yet accumulated many rare neighbours. Both approaches are powerful in principle — and both are dangerously sensitive to a flaw in their GWAS inputs.

BOX 6.1 · THE FLAW THAT MATTERS MOST

Population stratification, in one picture

Population stratification is a subtle correlation between ancestry and environment that contaminates GWAS effect sizes. Suppose people in the north of a region are slightly taller for environmental reasons (diet, say) *and* happen to carry certain ancestry-linked variants more often. A GWAS can then mistake those ancestry markers for “height genes.” Feed those contaminated effect sizes into a polygenic-selection test, and the ancestry gradient masquerades as a signal of selection. The variants aren't really affecting height, and selection isn't really acting — but the test lights up. Guarding against this is the single hardest problem in the recent-selection literature.

6.3 Case study: the height-selection episode

This is the cautionary tale every practitioner should carry, and it is why Mills & Mathieson strike such a careful tone. Around 2015–2016, several high-profile studies reported **polygenic selection on height** in Europe: northern Europeans appeared to have been selected for greater height, based on GWAS effect sizes from a large meta-analysis. It was an elegant, widely-cited result. Then in 2019, follow-up work using cleaner effect sizes — from the more uniform UK Biobank and from sibling-based designs that are immune to stratification — found that the signal **largely evaporated**. Most of the apparent “selection” had been residual population stratification leaking into the original effect sizes. A landmark finding was, to a large degree, a statistical artifact.

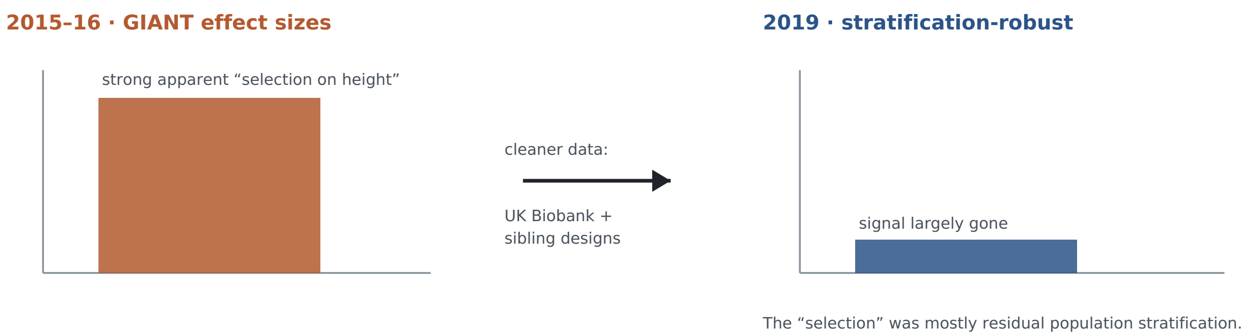

Figure 6.1 A landmark result that didn't hold. The height-selection signal reported from one set of GWAS effect sizes shrank to near nothing once stratification-robust data were used. The episode reset the field's confidence in polygenic-selection claims and is the backdrop to Mills & Mathieson's caution.

6.4 Ancient DNA: the cleaner path

The constructive message of Mills & Mathieson is that the surest way to study recent selection is to stop inferring past frequencies indirectly from modern genomes and instead *measure them directly*. **Ancient DNA** — genomes sequenced from archaeological remains

— lets you sample allele frequencies at multiple points in the past and simply watch a variant rise or fall through time. This sidesteps much of the demographic and stratification confounding that plagues modern-genome methods, and the rapidly growing ancient-genome record has made it a practical strategy. It is the cleanest test of recent selection we currently have.

> **THE LESSON TO CARRY FORWARD**
>
> Any claim of recent or polygenic selection will — correctly — be met with "is that selection, or is it stratification and demography?" Designing an analysis to answer that question *first*, rather than defending against it later, is what separates a durable result from the next retraction.

SUMMARY

- Polygenic traits are shaped by many tiny-effect variants; selection nudges all of them slightly and leaves no single sweep to detect.
- Polygenic-selection tests aggregate GWAS effect sizes to look for coordinated frequency shifts; SDS reads very recent change.
- These tests are acutely vulnerable to population stratification in the GWAS inputs.
- The 2015–2019 height episode showed a celebrated selection signal to be largely a stratification artifact.
- Ancient DNA measures past allele frequencies directly and is the cleanest current test of recent selection.

KEY TERMS

polygenic trait · GWAS · effect size · polygenic-selection test · singleton density score (SDS) · population stratification · sibling-based design · ancient DNA

CHECK YOUR UNDERSTANDING

1. Why do the haplotype and spectrum statistics of Chapter 4 struggle to detect selection on a highly polygenic trait?
2. Explain how population stratification can manufacture a false signal of polygenic selection.
3. What specific advantage does ancient DNA have over modern-genome inference for studying recent selection?

CHAPTER 7

Reading the Three Papers

THIS CHAPTER MAPS THE CONCEPTS ONTO YOUR READING, SO YOU KNOW WHAT TO EXTRACT FROM EACH PAPER.

You now have the scaffolding. Read the three papers in this order, with these goals.

Vitti, Grossman & Sabeti (2013) — “Detecting Natural Selection in Genomic Data”

This is your catalogue of footprints and statistics — the paper edition of Chapters 2–4. Read it to learn each test, what footprint it reads, and, above all, *what timescale it detects selection over*. As you go, build a one-page table with three columns: the statistic, the footprint (spectrum / haplotype / differentiation / cross-species), and its timescale. When you finish, that table is your working reference for the rest of the project. Don't get lost in the derivations; you want the map, not the algebra.

Cheng & Steinrücken (2024) — “Population Genomic Scans for Natural Selection and Demography”

This is the modern methods paper and the embodiment of Chapter 5. Read it as one sustained argument: *you cannot interpret a selection scan without a demographic model underneath it.* Pay attention to the coalescent framing, to the ancestral-recombination-graph tools (Relate, tsinfer, and allele-age/tree-based inference), and to how each method handles the demography-versus-selection problem. This is the paper that will most shape how you actually run analyses, so read it twice — once for the concepts, once for the tool names you'll need.

Mills & Mathieson (2022) — “The challenge of detecting recent natural selection in human populations”

This is Chapter 6 in the authors' own words, and it should shape how you *frame claims*, not just how you compute them. Read it for the failure modes: what breaks in recent and polygenic selection tests, why the height episode happened, and why ancient DNA and rigorous nulls matter. Let it install a healthy reflex — every time you find a signal, your first move should be to ask what non-selective process could have produced it.

HOW THE THREE FIT TOGETHER

Vitti gives you the tools. **Cheng & Steinrücken** tells you the null model those tools require. **Mills & Mathieson** tells you how the whole enterprise fails when you skip that step. Read in that order and each paper answers a question the previous one raised.

APPENDIX A

Databases and Data Access

Organized by how hard they are to get into. Start the slow ones immediately, because access approvals — not analysis — are usually the rate-limiting step in the first year.

Tier 0 — Open, use today (no application)

Fully public resources you can explore this week. Ideal for prototyping and for a first pass with precomputed statistics.

- ☐ **1000 Genomes Project** (via IGSR / EBI) — the reference panel of diverse human genomes on which most selection browsers are built.
- ☐ **1000 Genomes Selection Browser** and **PopHuman / PopHumanScan** — precomputed selection statistics across the genome; where a first-pass locus scan should live.
- ☐ **gnomAD** — population allele frequencies for essentially every variant; your quick reference for “how common is this, and where.”
- ☐ **GWAS Catalog** — curated trait–variant associations.
- ☐ **Ensembl / UCSC Genome Browser** — genome annotation, genes, regulatory features, conservation tracks.

Tier 1 — Light registration (summary data)

- ☐ **NIAGADS** (NIA Genetics of Alzheimer's Data Storage Site) — disease-GWAS summary statistics; some open, some by brief application.

Tier 2 — Controlled access via dbGaP (start the paperwork early)

Individual-level whole-genome data. Requires an eRA Commons account and a Data Access Request signed by your institution's signing official — involve your lab administrator and sponsored-programs office at the outset, as approval takes time.

- ☐ **TOPMed** (via dbGaP / BioData Catalyst) — a large, diverse whole-genome collection for replication.
- ☐ **Disease-specific sequencing resources** administered through dbGaP / NIAGADS — often as central to a disease-genetics project as TOPMed itself.

Tier 3 — All of Us Researcher Workbench (begin in month one)

A cloud-based controlled environment rather than a download: you analyze inside their workspace and pay for compute. The onboarding sequence — each step with its own queue — is:

- ☐ Register with institutional credentials and complete identity verification.
- ☐ Have your institution sign the Data Use Agreement.
- ☐ Complete the required responsible-conduct / CITI training.
- ☐ Gain Controlled Tier access to the whole-genome data linked to electronic health records, and budget for cloud compute.

> **THE NATURAL WORKFLOW**
>
> Prototype in the Tier 0 browsers → pull disease loci and summary statistics from Tier 1 → replicate in Tier 2 (individual-level whole genomes) → do the diversity-plus-phenotype work in Tier 3 (All of Us). The access difficulty rises exactly as the scientific payoff does, so begin the Tier 2–3 applications long before you need the data.

APPENDIX B

Glossary

Admixture
The mixing of two or more previously separated populations; can inflate F_{ST} and create long-range LD.

Allele age
How long ago a variant arose; estimable from genealogies and informative about selection.

Ancestral / derived allele
The original versus the newer version of a variant, assigned by comparison to an outgroup species.

Ancestral recombination graph (ARG)
The full set of genealogical trees along a chromosome, accounting for recombination.

Ancient DNA
DNA sequenced from archaeological remains, allowing allele frequencies to be measured directly at past time points.

Balancing selection
Selection that maintains multiple variants at intermediate frequency (e.g., heterozygote advantage; HLA diversity).

Bottleneck
A sharp temporary reduction in population size; purges diversity and lengthens haplotypes genome-wide.

Coalescent
A backward-in-time model tracing sampled lineages until they merge into common ancestors; the basis of modern inference.

Demography
A population's history of sizes, splits, and migrations; the principal confounder of selection detection.

Effective population size (N_e)
The size that determines the strength of drift; for humans about 10,000, far below the census size.

Effect size
The estimated contribution of a variant to a trait, from a GWAS.

EHH
Extended haplotype homozygosity; how far an identical haplotype extends around a core variant.

Fixation
Reaching a frequency of 100%; the endpoint of a completed sweep or of drift.

F_{ST}
A measure of how much variation at a site is due to differences between populations.

Genetic drift
Random change in allele frequency from the chance of reproduction; stronger in small populations.

GWAS
Genome-wide association study; scans the genome for variants statistically associated with a trait.

Haplotype
A specific combination of variants inherited together on one stretch of chromosome.

Hard / soft sweep
A sweep from a single new mutation (hard) versus from standing or repeated variation (soft, harder to detect).

Hitchhiking
The rise in frequency of variants linked to a favoured one.

iHS
Integrated haplotype score; flags variants on unusually long haplotypes within a population.

Linkage disequilibrium (LD)
Non-random association of nearby variants; the reason sweeps leave neighbourhood-wide footprints.

Neutral theory
The proposition that most molecular variation is neutral and governed by drift; the field's null hypothesis.

Polygenic trait
A trait shaped by many variants of small effect; the setting where selection is hardest to detect.

Pooling
The second pass in building an SFS: tallying the per-site variant counts to find how many sites fall in each frequency class, so individual sites lose their identity.

Population stratification
Correlation between ancestry and environment that biases GWAS effect sizes and can fake selection signals.

Positive (directional) selection
Selection that drives a beneficial variant up in frequency; produces sweeps.

Purifying (negative) selection
Selection that removes harmful variants; the cause of sequence conservation.

Recombination
The per-generation shuffling of chromosomal segments; erodes swept haplotypes over time.

Selective sweep
The rise of a favoured variant and its neighbourhood; leaves reduced diversity and long haplotypes.

Singleton
A variant seen in exactly one chromosome of a sample — the rarest possible class and the leftmost bin of the SFS.

Singleton density score (SDS)
A statistic reading very recent frequency change from the density of the rarest nearby variants.

Site frequency spectrum (SFS)
The histogram of how common variants are across a region; the workhorse summary of the data.

Tajima's D
A statistic whose sign reports the shape of the SFS: negative for sweeps/expansions, positive for balancing/contractions.

APPENDIX C

Further Reading

The three companion papers

Vitti JJ, Grossman SR, Sabeti PC. Detecting Natural Selection in Genomic Data. *Annual Review of Genetics*. 2013;47:97–120.

Cheng X, Steinrücken M. Population Genomic Scans for Natural Selection and Demography. *Annual Review of Genetics*. 2024;58:319–339.

Mills MC, Mathieson I. The challenge of detecting recent natural selection in human populations. *Proceedings of the National Academy of Sciences*. 2022;119(15):e2203237119.

Foundational and background

Kimura M. Evolutionary rate at the molecular level. *Nature*. 1968;217:624–626. (The original neutral-theory paper.)

Hartl DL, Clark AG. *Principles of Population Genetics*. Sinauer. (A standard textbook for deeper treatment of any chapter here.)

Coop G. *Population and Quantitative Genetics* (open-access lecture notes). (An accessible, freely available course-level treatment.)

Nielsen R. Molecular signatures of natural selection. *Annual Review of Genetics*. 2005;39:197–218. (A widely cited earlier review, useful alongside Vitti.)

On the cautionary case

Follow-up analyses in 2019 (using UK Biobank and sibling-based designs) that reassessed the earlier reports of polygenic selection on height; see the references within Mills & Mathieson (2022) for the specific studies.

Prepared as a study companion. Figures are schematic and drawn for teaching, not to scale.